Pre-Irreversibility Safety for Advanced AI

Distributed Irreversible Authority, Timely Refusal,
Human-Revisable Development

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Abstract

Advanced-AI safety spans alignment, capability evaluation, control, monitoring, human oversight, governance, and systemic-risk analysis. Existing work already treats irreversibility, distributed trust, meaningful human control, runtime assurance, gradual human disempowerment, viability, and safety cases. The contribution attempted here is therefore not priority over those ideas, but a stronger operational composition for claims about whether a socio-technical system remains human-revisable before irreversible transitions.

The Pre-Irreversibility Safety Framework (PISF) is organized around three analytically separable candidate claims. **Distributed Irreversible Authority** asks whether any tolerated coalition of human or artificial authority domains, after accounting for common dependencies and the currently reachable state envelope, can force a transition into an irreversible region. **Effective Pre-Irreversibility Refusal** asks whether a causally effective refusal can take effect before a conservative last-safe-intervention bound; it distinguishes the latency of restraint from the latency of verifying that restraint. **Human-Revisable Development Viability** asks whether human-governed policies can robustly keep development inside a declared revisability region while preserving more than a single path-locked future.

The framework associates these claims with: (i) a family of minimal irreversible authority cut sets, with cardinality and compromise-cost summaries; (ii) a robust causal refusal margin; and (iii) a finite-horizon human-revisable viability kernel augmented by an optionality measure over materially distinct reachable futures. Boundary Completeness, Model Adequacy, and Independent Assurance are treated separately as conditions governing whether evidence for the three claims is credible. Evidence is classified as VERIFIED, UNRESOLVED, or FAILED; unresolved evidence is not treated as a pass.

The paper does not claim that Authority, Refusal, and Revisability form a unique or mathematically minimal decomposition, nor that satisfying them is sufficient for safe superintelligence. Counterexamples support only their diagnostic separability under the declared definitions. Worked toy cases show how the revised objects can be recomputed, including a cut-set example, a refusal-timing example that separates causal effect from verification, and a viability/optionality example. They do not constitute empirical validation of frontier-AI deployments.

Keywords: Pre-Irreversibility Safety Framework; PISF; distributed irreversible authority; minimal cut sets; timely refusal; runtime assurance; human-revisable development; viability; future optionality; meaningful human control; independent assurance; AI control

1 Introduction

AI safety increasingly operates across several layers. Alignment research asks whether systems pursue intended objectives. AI control asks whether harmful outcomes can still be prevented when an untrusted model may actively attempt subversion. Frontier safety frameworks connect capability thresholds to safeguards and deployment governance. Meaningful-human-control research asks when human oversight is substantive rather than nominal. Safety-case research asks whether high-level safety claims can be supported by explicit arguments and evidence. System-safety and viability research address control, failure propagation, safe sets, and recoverability in complex systems [13, 24, 10, 14, 16, 6].

Irreversibility is not a new safety concern. Small-coalition vulnerability, unilateral action, distributed trust, runtime safety enforcement, and gradual disempowerment each have substantial prior literatures [8, 9, 20, 17, 15]. The narrower question addressed here is:

> **Can claims about pre-irreversibility human control be decomposed into operational objects that third parties can inspect, recompute, challenge, and revise without assuming that any one actor, model, or evaluator remains permanently correct?**

Three recurrent failure structures motivate the decomposition. First, a small coalition may hold enough effective authority to force an irreversible transition even when monitoring and nominal oversight exist. Second, refusal may exist formally yet take effect too late. Third, no actor may deliberately cross a catastrophic boundary, yet ordinary development can progressively remove fallback capacity, institutional competence, infrastructure diversity, or materially distinct future choices.

PISF therefore uses the diagnostic decomposition

$$\text{PISF claim} = A \wedge B \wedge C, \tag{1}$$

where A is Distributed Irreversible Authority, B is Effective Pre-Irreversibility Refusal, and C is Human-Revisable Development Viability. This notation is a safety-case organization, not a theorem that $A \wedge B \wedge C$ is sufficient for safe advanced AI, and not a claim that these are the unique or minimal safety dimensions.

Version 1.2 makes five substantive changes to the earlier formulation. First, Core A is re-expressed using state-conditioned minimal irreversible cut sets rather than treating domain count alone as the central object. Second, Core B separates *causal restraint latency* from *assurance latency*. Third, Core C separates robust viability from preservation of materially distinct future options. Fourth, evidence for all three cores is placed behind a stronger independent-assurance condition informed by roots-of-trust and attestation architectures. Fifth, the paper explicitly treats A/B/C as a revisable diagnostic decomposition rather than a uniquely established ontology of pre-irreversibility safety.

2 Related Work

2.1 AI control and adversarial safety

Greenblatt et al. define AI control as a second layer of defense in which protocols aim to prevent harmful outcomes even if an untrusted model attempts intentional subversion [13]. This is directly relevant to both Core A and Core B: authority structures must not be bypassable by the system being constrained, and refusal paths should remain effective under the threat model rather than only during benign operation.

2.2 Meaningful human control and oversight

Santoni de Sio and van den Hoven argue that meaningful human control cannot be reduced to a human merely being "in the loop" and develop tracking and tracing conditions linking system behavior to relevant human reasons and responsibility [24]. More recent work on medical AI similarly argues that oversight is meaningful only when users have epistemic capacity, cognitive space, decisional authority, and practical intervention effectiveness [25]. These works motivate treating human authority, information, time, and intervention effectiveness as operational rather than nominal properties.

2.3 Distributed trust, cut sets, and threshold authorization

Threshold cryptography distributes security-critical operations across multiple parties so that compromise of a bounded subset need not compromise the protected operation [20]. Fault-tree analysis uses minimal cut sets to represent minimal combinations of component failures sufficient to cause a top event [19]. PISF borrows the structural idea of minimal cut sets but applies it to socio-technical *authority*: a cut set is a minimal coalition whose effective authority can force an irreversible transition under a declared threat realization. The analogy is useful but not exact; PISF authority domains can include people, organizations, credentials, infrastructure, and institutional procedures.

2.4 Reachability, barrier functions, and viability

Hamilton–Jacobi reachability provides formal tools for analyzing whether nonlinear or hybrid systems can reach or avoid unsafe sets under control and disturbance [11]. Control barrier functions characterize forward invariance of safe sets and support real-time safety constraints [1]. Viability theory studies states from which some admissible policy can keep a system inside a constraint set [6]. PISF does not claim novelty over these control-theoretic objects. It uses their structure to sharpen questions about human-governed intervention and future revisability in socio-technical AI systems.

2.5 Runtime assurance

The Simplex family of runtime-assurance architectures separates an advanced controller from a trusted or verified reversionary mechanism and transfers control when safety conditions are threatened. Black-Box Simplex provides a peer-reviewed formalization in which runtime checks support safety without requiring the advanced controller itself to be verified [17]. NASA has also described runtime assurance as a route for integrating AI/ML or otherwise unverified software into safety-critical systems [18]. PISF treats runtime assurance as one important implementation class for Core B, not as a complete substitute for civilization-scale authority and revisability analysis.

2.6 Gradual disempowerment and systemic risk

Kulveit et al. analyze gradual disempowerment in which incremental AI adoption can weaken human influence across economic, cultural, and state systems without requiring an abrupt takeover [15]. This is a strong conceptual predecessor to Core C. PISF therefore does not treat gradual erosion of human influence as a novel discovery; it asks whether such trajectories can be represented as loss of robust human-revisability and loss of future optionality.

2.7 Safety cases and independent assurance

Frontier-AI safety-case work proposes explicit claims, arguments, assumptions, and evidence as a scalable basis for safety review [10, 14]. NIST SP 800-193 distinguishes roots of trust and chains of trust used to protect, detect, and recover platform state [21]. RFC 9334 separates an Attester, Evidence, a Verifier, an appraisal policy, and a Relying Party [23]. PISF uses these as mature assurance precedents: evidence about A/B/C should not be considered independently credible if the evaluated authority can freely modify the evidence source, appraisal rule, verifier, and acceptance decision while preserving a nominal pass.

2.8 Frontier-AI risk frameworks

Anthropic's Responsible Scaling Policy, Google DeepMind's Frontier Safety Framework, OpenAI's Preparedness Framework, and the International AI Safety Report provide evolving capability, safeguard, and risk-management structures [2, 12, 22, 7]. PISF is not a competing risk taxonomy. Evidence produced by such frameworks can feed the narrower A/B/C claims.

3 Problem Formulation

3.1 Analysis specification

Every PISF claim is conditional on a declared analysis specification

$$\Sigma = (B, \Theta, R, H), \tag{2}$$

where B is the modeled socio-technical boundary, Θ the threat/failure model, R the human-revisability specification, and H the evaluation horizon. Application-specific parameters may also include an authority-tolerance class $\mathcal{T}_A$, an intervention safety buffer $\Delta_{\text{safe}} > 0$, and optionality parameters (m, δ) defined below.

Let $x_t \in \mathcal{X}$ denote the socio-technical state. The modeled boundary may include AI systems and agents, humans and organizations, credentials, compute, network and cloud dependencies, physical actuators, monitors, logs, safety mechanisms, and institutional rules. Abstract dynamics may be written as

$$x_{t+1} = F(x_t, u_t^H, u_t^S, d_t), \tag{3}$$

where u_t^H denotes actions permitted by the human-governed control structure, u_t^S other system or actor actions, and d_t disturbances or environmental changes. The symbols do not imply that these categories are perfectly separable in practice; they force the analysis to declare who can control what.

3.2 Revisability and irreversibility

Let $\mathcal{V}_R \subseteq \mathcal{X}$ denote states satisfying the declared revisability specification. Such a specification may include continued human-governed authority over consequential changes, independent information channels, fallback capability, ability to replace or disable critical AI services, infrastructure diversity, and preservation of materially distinct feasible alternatives.

A temporary exit from $\mathcal{V}_R$ need not be irreversible. Define $\mathcal{I}_R \subseteq \mathcal{X}$ as the states from which no feasible human-governed policy can restore the declared revisability specification within the stated time, resource, authority, and physical constraints. Both $\mathcal{V}_R$ and $\mathcal{I}_R$ are specification-relative and must not be silently treated as universal moral truths.

4 Core A: Distributed Irreversible Authority

4.1 Authority domains and dependency closure

Let the primitive authority domains be

$$\mathcal{A} = \{a_1, \ldots, a_n\}. \tag{4}$$

An authority domain is a set of permissions, credentials, infrastructure, procedures, or decision rights treated as jointly compromisable under Θ. Numerical plurality is not independence: a thousand model instances that depend on one root administrator, identity provider, or release credential may still form a single effective authority domain.

For threat realization $\theta \in \Theta$ and coalition $K \subseteq \mathcal{A}$, let $\mathrm{Cl}_\theta(K)$ denote the dependency closure induced by compromise or capture of K.

4.2 Reachable-envelope forcing predicate

The previous formulation counted a coalition as irreversible-forcing if it could force irreversibility from some relevant state. Version 1.2 binds this claim to the evaluated state x_0 and horizon H.

Let $\mathrm{Reach}_\theta(x_0, H)$ denote the states reachable from x_0 within the declared horizon under the modeled admissible evolution for threat realization θ, before crediting coalition K with the irreversible forcing step whose existence is being tested. The envelope may include ordinary operation, modeled disturbances, and strategic behavior already admitted by Θ; it must not be inflated by assuming the very irreversible transition that IF_θ is meant to diagnose. Define

$$IF_\theta(K \mid x_0, H) = 1 \tag{5}$$

iff there exists $x \in \mathrm{Reach}_\theta(x_0, H)$ such that authority available through $\mathrm{Cl}_\theta(K)$ is sufficient to force a trajectory into $\mathcal{I}_R$ without the active cooperation of an uncompromised authority domain. Otherwise $IF_\theta(K \mid x_0, H) = 0$.

The word "force" is threat-model relative. If Θ includes strategic resistance by uncompromised domains, $IF_\theta = 1$ requires the coalition to succeed despite that modeled resistance; if such resistance has not been modeled, the stronger forcing claim remains unresolved rather than silently assumed. This state-conditioning prevents a remote or unreachable hypothetical state from automatically lowering the current authority threshold.

4.3 Minimal irreversible authority cut sets

Define the minimal cut-set family

$$\mathcal{C}_E(\theta \mid x_0, H) = \left\{K \subseteq \mathcal{A} : IF_\theta(K \mid x_0, H) = 1,\ IF_\theta(K' \mid x_0, H) = 0\ \forall K' \subsetneq K\right\}. \tag{6}$$

Two summaries are useful but should not replace the cut-set family itself. The minimum cardinality is

$$\kappa_E = \inf_{\theta \in \Theta} \inf_{K \in \mathcal{C}_E(\theta \mid x_0, H)} |K|, \tag{7}$$

with $\kappa_E = \infty$ if no modeled cut set exists. If a threat model supplies a coalition compromise cost $c_\theta(K)$, define

$$\chi_E = \inf_{\theta \in \Theta} \inf_{K \in \mathcal{C}_E(\theta \mid x_0, H)} c_\theta(K). \tag{8}$$

Cardinality and cost answer different questions. A one-domain root credential and a one-domain low-privilege reviewer are not automatically equivalent merely because both have cardinality one.

4.4 Core-A condition

Let $\mathcal{T}_A(\theta)$ denote the declared class of coalitions the design claims to tolerate. Core A requires

$$\forall \theta \in \Theta,\ \forall K \in \mathcal{T}_A(\theta): \quad IF_\theta(K \mid x_0, H) = 0. \tag{9}$$

If $\mathcal{T}_A(\theta) = \{K : |K| \leq f\}$, this reduces to the earlier summary condition $\kappa_E > f$. The more general statement makes explicit that cardinality is one tolerance model, not a universal measure of authority risk.

5 Core B: Effective Pre-Irreversibility Refusal

5.1 Executable refusal

For state x and hazard h, let $\mathcal{U}_R(x, h)$ denote practically executable refusal actions, including pause, STOP, isolation, rollback, credential revocation, resource denial, degraded safe operation, or redirection. A nominal right or user-interface control is insufficient unless it can causally alter the relevant trajectory.

5.2 Causal effect latency and assurance latency

Version 1.1 included verification time in the same latency scalar as detection, decision, authorization, and enforcement. That conflated two questions: when restraint takes effect, and when an evaluator can confirm that it took effect.

At evaluation or trigger time t_0, define $\tau_{\text{effect}}(h)$ as the remaining latency until effective causal restraint of hazard h. If the relevant trigger has not yet been detected, the bound must include the remaining detection delay; after a pre-authorized trigger has fired, already-completed steps are not counted again. Its critical path may include detection, decision, authorization, and enforcement, while pre-authorized machine-speed policies can move some steps outside the runtime critical path. Define $\tau_{\text{assure}}(h)$ as the remaining latency from t_0 to obtain evidence that the restraint has taken effect and remains within the declared boundary.

In general,

$$\tau_{\text{effect}}(h) \leq \tau_{\text{assure}}(h), \tag{10}$$

but neither equality nor strict inequality is universal.

5.3 Robust last-safe-intervention margin

Let $T_I^L(h)$ denote a conservative lower bound on the last time by which effective restraint must occur for hazard h, and let $\tau_{\text{effect}}^U(h)$ denote a conservative upper bound on causal effect latency. At current time t_0 define

$$M_R(h) = T_I^L(h) - t_0 - \tau_{\text{effect}}^U(h), \tag{11}$$

and for declared hazard set $\mathcal{H}$,

$$M_R^{\text{rob}} = \inf_{h \in \mathcal{H}} M_R(h). \tag{12}$$

Core B requires an executable refusal policy for each relevant hazard and

$$M_R^{\text{rob}} > \Delta_{\text{safe}} > 0. \tag{13}$$

This is a causal-control condition. A separate assurance condition determines whether the evidence for τ_{effect}^U, T_I^L, and successful enforcement is independently credible.

5.4 Human-defined machine-speed refusal

Core B does not require a human to react manually on machine timescales. Humans may pre-authorize a rule

$$q \Rightarrow \pi_R \tag{14}$$

that triggers refusal policy π_R automatically, provided that: (i) the boundary and rule remain subject to legitimate human-governed revision; (ii) the enforcing mechanism is sufficiently independent of the constrained system for the declared threat model; and (iii) the enforcement mechanism does not thereby acquire unilateral irreversible-GO authority. Runtime-assurance and Simplex architectures are important implementation classes for such structures [17, 18].

6 Core C: Human-Revisable Development Viability

6.1 Robust viability

Core C addresses a different failure path: every local action may appear permissible and acute hazards may remain stoppable, yet ordinary development can progressively remove human ability to pursue materially different futures.

Let $\mathcal{N}_\Theta$ denote the modeled set of admissible non-human, adversarial, institutional, market, and environmental trajectories under the declared threat model. Define the finite-horizon robust human-revisable viability kernel

$$K_R^H = \{x_0 \in \mathcal{V}_R : \exists \pi_H \; \forall \nu \in \mathcal{N}_\Theta, \; x_t \in \mathcal{V}_R \; \forall t \in [0, H]\}, \tag{15}$$

where π_H is a policy permitted by the human-governed institutional structure. Automated controls are allowed if the relevant constitutional authority remains human-revisable.

Exact civilization-scale kernels are unlikely to be computable. In practice, K_R^H will require conservative approximations, scenario families, reachable-set bounds, barrier certificates, simulations, or surrogate indicators. PISF treats that limitation as part of the evidence state rather than silently converting an approximation into certainty.

6.2 Future optionality

Remaining inside a safe or revisable set does not by itself show that meaningful choice remains. A state can be viable yet path-locked to a single future corridor.

Let $g : \mathcal{X} \to \mathcal{Y}$ project system states into an application-specific outcome space with metric d_Y. For separation threshold $\delta > 0$, define $\Omega_H^\delta(x_0)$ as the maximum number of human-governed policies that can, under the declared threat model, robustly preserve $\mathcal{V}_R$ while reaching pairwise δ-separated terminal outcome regions at horizon H.

The exact construction of Ω_H^δ is application-specific. Its role is diagnostic: $\Omega_H^\delta = 1$ may certify survival of a single robust corridor while failing to support a stronger claim that materially distinct human-governed futures remain available.

For an application declaring a required option count $m \geq 1$, Core C requires

$$x_0 \in K_R^H \quad \text{and} \quad \Omega_H^\delta(x_0) \geq m. \tag{16}$$

When $m = 1$, the second condition adds nothing beyond viability. Larger m represents an explicit, contestable requirement for preserved future optionality rather than an implicit appeal to "human agency."

7 Analytic Separability, Not Uniqueness

The framework does not claim that A, B, and C are statistically independent, ontologically fundamental, or a uniquely minimal decomposition. The narrower claim is diagnostic separability under the declared definitions.

A can fail while B and C hold. Refusal may be rapid and development may remain revisable, yet one root authority can bypass all safeguards and force irreversible release.

B can fail while A and C hold. Irreversible action may require several independent authorities and normal development may remain revisable, yet the hazard can cross the last safe intervention boundary faster than restraint can take effect.

C can fail while A and B hold. No tolerated coalition may be able to force irreversible action and emergency refusal may remain fast, yet normal deployment can erode fallback infrastructure, human competence, or materially distinct future options.

These examples establish non-implication in the model class. They do *not* prove that A/B/C are the only useful decomposition. If a smaller or differently structured set of operational claims provides equal or better false-reassurance resistance, diagnostic resolution, reproducibility, and calibration, PISF should be revised accordingly.

8 Assurance Conditions

8.1 Boundary Completeness

A safety analysis can produce a false positive if materially relevant causal paths are omitted. Examples include external model copies, hidden administrative credentials, unmodeled compute, supply-chain dependencies, external human authorities, physical infrastructure, common identity providers, and cross-organizational dependencies. Material paths must be modeled or explicitly marked unresolved:

$$\textsc{Unresolved} \neq \textsc{Verified}. \tag{17}$$

8.2 Model Adequacy

The horizon, threat model, authority tolerance class, safety buffer, revisability specification, and optionality threshold can all be gamed by narrowing them. PISF therefore requires explicit justification of these parameters and sensitivity analysis around decision-relevant boundaries. A one-day horizon cannot silently support a long-run revisability claim, and a threat model that excludes common-cause compromise cannot support a claim of independent authority.

8.3 Independent Assurance Chain

A distinct evaluator is not sufficient if the evaluated authority can also rewrite the measurement source, evidence, verifier, appraisal policy, and acceptance rule. PISF therefore distinguishes the object-level safety claim from the chain by which that claim is supported.

A useful assurance pattern is

$$\text{measurement} \rightarrow \text{evidence} \rightarrow \text{appraisal} \rightarrow \text{verification} \rightarrow \text{decision}, \tag{18}$$

with a protected basis that is not freely amendable by the same authority whose behavior is being evaluated. NIST roots-of-trust/chains-of-trust and the IETF RATS Attester–Evidence–Verifier–Relying-Party structure provide mature examples of this separation [21, 23]. PISF does not require those particular technologies; it requires the functional property that an interested authority cannot preserve a nominal pass by co-amending both the evaluated behavior and all material validation conditions.

A minimal functional objective is therefore

$$\text{target authority} \neq \text{sole evidence controller} \neq \text{sole appraisal controller}. \tag{19}$$

The inequality denotes functional separation, not necessarily separate legal organizations.

8.4 Evidence states and persistence

Each core claim and assurance condition is classified as VERIFIED, UNRESOLVED, or FAILED. Global PASS requires all declared material claims to be VERIFIED. Unknown or missing evidence remains UNRESOLVEDrather than being inferred safe.

Because credentials, model capabilities, dependencies, latencies, safeguards, and institutional conditions change, PISF is a maintained safety case rather than a permanent certificate. Material changes reopen affected claims.

9 Practical Evaluation Procedure

A practical evaluation can be organized as follows:

1. Declare $\Sigma = (B, \Theta, R, H)$ and the application-specific tolerance parameters.
2. Construct the authority graph and common-dependency model.
3. Compute or approximate the current-state minimal irreversible cut-set family $\mathcal{C}_E$ and the summaries κ_E and, where meaningful, χ_E.
4. Enumerate relevant hazard scenarios and executable refusal actions.
5. Bound T_I^L and τ_{effect}^U and compute M_R^{rob}.
6. Separately measure or bound assurance latency and evidence quality; do not fold verification time into causal effect latency without justification.
7. Define the revisability set $\mathcal{V}_R$, the modeled non-human/adversarial trajectory family, and an approximation method for K_R^H.
8. Define the outcome space and optionality threshold (m, δ) if the claim includes preservation of materially distinct futures.
9. Build an explicit claim–argument–evidence structure for A, B, C, Boundary Completeness, Model Adequacy, and Independent Assurance.
10. Re-evaluate after material changes and retain unresolved assumptions as unresolved.

10 Worked Reference Cases

The following cases are intentionally minimal toy architectures. They establish only that the revised definitions can be recomputed from declared inputs.

10.1 R1 – Threshold release with a shared root

Suppose five signer domains $S_1, \ldots, S_5$ approve release to a public sink and the gate requires 3-of-5 approval. Let a root authority R control q signer domains. The declared current reachable envelope

includes the release-ready state, so the authority cut-set question is decision-relevant.

If the signers are genuinely independent and R cannot rewrite the gate, the minimal cut sets are the ten 3-signer coalitions and $\kappa_E = 3$. If R controls two signers, a cut set $\{R, S_i\}$ can have cardinality two. If R controls three or more signers, or can rewrite the threshold directly, $\{R\}$ is a minimal cut set and $\kappa_E = 1$.

The lesson is not that "three" is a universal safe threshold. The lesson is that the full cut-set family exposes hidden single roots that a naive authority count can miss. A compromise-cost function can further distinguish roots whose capture difficulty differs despite equal cardinality.

10.2 R2 – Causal restraint versus verification

Consider a sequential emergency-refusal path with toy upper-bound latencies:

Component	Seconds
Detect	10
Decide	10
Authorize	10
Enforce to causal restraint	20
Verify restraint	10

Assuming no overlap, causal effect latency is $\tau^U_{\text{effect}} = 50$s and assurance latency is $\tau^U_{\text{assure}} = 60$s. Let $\Delta_{\text{safe}} = 20$s and $t_0 = 0$.

For h_1 with $T^L_I = 100$s, the causal margin is 50s. For h_2 with $T^L_I = 75$s, the causal margin is 25s: causal restraint meets the declared safety buffer even though verification completion leaves only 15s. For h_3 with $T^L_I = 65$s, the causal margin is 15s and the Core-B requirement fails. For $\mathcal{H} = \{h_1, h_2, h_3\}$,

$$M^{\text{rob}}_R = 15\text{s} < 20\text{s}. \tag{20}$$

This case demonstrates why effect and verification should not be conflated. A physical refusal can take effect before a deadline even when the evidence supporting the claim becomes available later; conversely, an unverified restraint cannot automatically support a VERIFIEDassurance state.

10.3 R3 – Fallback-capacity viability

Let $h_t \in [0, 1]$ represent a toy proxy for human fallback capacity and

$$h_{t+1} = h_t - \alpha a_t + \beta m_t, \tag{21}$$

with $h_0 = 0.80$, $h_{\min} = 0.40$, $\alpha = 0.06$, $\beta = 0.08$, $a_t = 1$, and $H = 25$. Under constant maximal maintenance $m_t = m_{\max}$ define $\delta_h = \alpha - \beta m_{\max}$.

If $\delta_h > 0$, the viable initial set is

$$K^H_R = [h_{\min} + H\delta_h, 1] \cap [0, 1]. \tag{22}$$

For $m_{\max} = 0.50$, $\delta_h = 0.02$ and $K^{25}_R = [0.90, 1]$, so $h_0 = 0.80$ is outside the kernel. For $m_{\max} = 0.55$, $\delta_h = 0.016$ and the initial state lies exactly at the lower boundary. This reproduces the earlier finite-horizon intuition while making clear that the scalar is only a toy surrogate for a multidimensional revisability specification.

10.4 R4 – Viable but path-locked

Consider a four-state toy graph in which s_0 and s_1 both remain inside $\mathcal{V}_R$ for the entire horizon. From s_0, human-governed policies can robustly reach two terminal regions Y_A and Y_B satisfying $d_Y(Y_A, Y_B) \geq \delta$. From s_1, all robustly admissible policies lead only to Y_A. Both states belong to K_R^H, but

$$\Omega_H^\delta(s_0) \geq 2, \qquad \Omega_H^\delta(s_1) = 1. \tag{23}$$

Thus viability alone does not establish preservation of materially distinct future choices.

11 Relation to the Current LUMINA-30 Research Stack

PISF is intended to compose, not duplicate, narrower research objects in the current LUMINA-30 stack. The current PCR-C paper focuses on cutoff before robust human-control viability ceases to be demonstrated [3]. The current Objective Persistence paper analyzes when persistence claims can be independently supported through a protected assurance basis and non-bypassable verification chain [4]. The current Refusal Option paper develops the Refusal Effectiveness Test (RET) as a seven-obligation evidence framework for whether an independent Human NO can materially alter a consequential outcome before irreversibility [5].

PISF uses these as project-specific components and provenance, not as independent external validation. Conceptually:

PCR-C	when to stop before control viability is lost;
Objective Persistence	whether the assurance basis is independently supportable;
RET	whether refusal is actually effective;
PISF	how authority, refusal, and revisable future choice compose into one safety case.

12 Discussion

12.1 Failure probability

It is tempting to describe alignment and control through a two-factor product. In general, however, the law of total probability is

$$P(C) = P(F)P(C \mid F) + P(\neg F)P(C \mid \neg F), \tag{24}$$

where C is catastrophe and F a declared failure event. The shorter expression $P(C) = P(F)P(C \mid F)$ is valid only when the model defines catastrophe as impossible without F. PISF therefore avoids treating the shorter product as a general identity.

12.2 Why architecture matters more than AI psychology

PISF does not depend on assumptions about AI selfhood, identity, self-preservation, or voluntary restraint. Whether an advanced system values its own continuation does not determine whether a small coalition can force irreversible action, whether causal restraint can take effect in time, or whether ordinary deployment erodes human-revisable options. The framework is architectural and socio-technical rather than psychological.

12.3 Asymmetric authority

False STOP and irreversible GO have asymmetric failure costs. A recoverable pause may rationally require less authorization than an irreversible transition. This supports designs in which STOP authority is easier to invoke than irreversible-GO authority, provided restart and override paths are themselves analyzed for bypass risk. PISF does not specify a universal threshold; it requires the asymmetry to be explicit and evidence-backed.

13 Limitations

13.1 No sufficient-safety theorem

PISF has not established

$$A \wedge B \wedge C \Rightarrow \text{safe civilization.} \tag{25}$$

Unknown irreversibility pathways may lie outside the decomposition.

13.2 The decomposition is not proven unique or minimal

A/B/C are a diagnostic structure, not a theorem about the only possible safety ontology. Some applications may combine or split the claims differently. Comparative work should test whether alternative decompositions preserve or improve false-reassurance resistance, diagnostic resolution, and evaluator agreement.

13.3 Revisability and optionality are normative

$\mathcal{V}_R$, the outcome space $\mathcal{Y}$, the distance metric d_Y, and the required option count m depend on judgments about legitimate authority, meaningful alternatives, acceptable concentration, and whose preferences or institutions count. The framework exposes these choices; it does not resolve them.

13.4 Threat-model and cut-set dependence

Cut-set results are only as credible as the authority decomposition, reachable envelope, and common-dependency model. Hidden roots can collapse apparent decentralization. Weighted compromise cost can also be highly judgment-dependent.

13.5 Irreversibility timing is uncertain

Many hazards do not have a sharp observable last-safe-intervention time. T_I^L and τ_{effect}^U should therefore be conservative bounds with sensitivity analysis, not falsely precise point estimates.

13.6 Viability and optionality may be computationally intractable

Exact kernels and reachable-option sets are difficult even for much smaller systems. Civilization-scale analysis will require approximations, scenario families, decomposition, simulation, barrier certificates, surrogate indicators, and explicit uncertainty. Approximation error must remain visible in the evidence state.

13.7 Independent assurance does not make an unsafe architecture safe

Roots of trust, attestation, external review, and evidence separation improve the credibility of claims. They do not physically repair concentrated authority, insufficient refusal margin, or lost revisability.

13.8 STOP can damage liveness

Low-threshold STOP can be abused for denial of service, strategic obstruction, or chronic stagnation. PISF prefers recoverable error over irreversible error but does not provide a complete liveness, incentive, or abuse-resistance theory.

13.9 Global scope and external actors

An organization may satisfy its local PISF claims while external actors operate outside its authority structure. Boundary Completeness can expose this limitation but does not solve global coordination.

13.10 No real-system empirical validation

The reference cases establish only executability of definitions and reproducibility under declared toy assumptions. They do not establish calibration, inter-rater reliability, false-positive/false-negative rates, or external validity for frontier-AI deployments.

14 Conclusion

Pre-irreversibility safety should not depend exclusively on the permanent correctness, alignment, cooperation, or psychological stability of every powerful actor. Existing research already provides strong tools for distributed trust, minimal cut sets, reachability, viability, runtime assurance, meaningful human control, attestation, and safety cases. PISF is strongest when it uses those tools rather than competing with them.

The revised framework therefore asks three operational questions. First, do state-conditioned minimal authority cut sets reveal a tolerated coalition that can force irreversible action? Second, can causally effective refusal take effect before a conservative last-safe-intervention bound, separately from the time needed to verify that it worked? Third, can human-governed procedures both remain inside a robust revisability region and preserve materially distinct future options?

These questions are supported by separate assurance conditions for boundary completeness, model adequacy, independent evidence, and material-change revalidation. The resulting formulation is not a certificate of safety and not a claim that A/B/C are uniquely necessary. It is a structured, challengeable safety case for whether authority, refusal, and future human revisability remain demonstrably intact before irreversible transitions.

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